THE RANKERS ACADEMY — NAGPUR

MATHEMATICS TEACHING PLAN | 95 Content Lectures | DPP 1–95 | Each Lecture: 45 Minutes

1 Mathematics lecture per teaching day (Mon–Fri) | Saturday = DPP discussion + doubt session

Chapter 1: Sets, Relations & Functions | Apr 27 – May 4 | 6 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L1	Sets – types & operations	Types of sets; union, intersection, complement, difference; De Morgan's laws; Venn diagram problems	DPP 1	45 min
L2	Relations – types	Cartesian product; types of relations (reflexive, symmetric, transitive, equivalence); closure property	DPP 2	45 min
L3	Functions – types	Function definition; one-one, onto, bijective; onto function counting; composite functions; inverse function	DPP 3	45 min
L4	Domain & range	Domain of √, log, rational, modulus; range finding methods; greatest integer, fractional part functions	DPP 4	45 min
L5	Even-odd & periodic functions	Even/odd identification; periodic function T definition; period of trig functions; composition properties	DPP 5	45 min
L6	Functions PYQ drill	JEE Main 2019-2025 function PYQs; graph-based questions; number of functions/relations problems	DPP 6	45 min

Chapter 2: Trigonometry | May 5 – May 20 | 12 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L7	Trigonometric ratios & exact values	Six ratios; exact values at 0°,30°,45°,60°,90°,120°,150°,180°,270°,360°; signs in quadrants; ASTC rule	DPP 7	45 min
L8	Fundamental identities & allied angles	Pythagorean identities; sin(90°±x), cos(90°±x) etc.; sin(180°±x); sin(270°±x); sin(360°±x)	DPP 8	45 min
L9	Compound angle formulas	sin(A+B), cos(A+B), tan(A+B) derivations; all combinations; sin(A−B) etc.; product as sum	DPP 9	45 min
L10	Multiple & sub-multiple angles	sin2A, cos2A, tan2A derivation; sin3A, cos3A; tan3A; $\sin^2 A$ & $\cos^2 A$ in terms of cos2A	DPP 10	45 min
L11	Product-to-sum & sum-to-product	sinC+sinD = 2sin((C+D)/2)cos((C−D)/2); all 4 formulas; product formulas; applications	DPP 11	45 min
L12	Trigonometric equations	General solution of sinx=a, cosx=a, tanx=a; principal value; conditions; combined equations	DPP 12	45 min
L13	Complex trig equations	Equations reducible to quadratic; conditional identities; solving for x in [0, 2π]; number of solutions	DPP 13	45 min
L14	Inverse trigonometry – basics	Domain, range, principal values of $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$, $\cot^{-1}$, $\sec^{-1}$, $\operatorname{cosec}^{-1}$	DPP 14	45 min
L15	Inverse trig identities	$\sin^{-1}+\cos^{-1}=\pi/2$; $\tan^{-1}+\cot^{-1}=\pi/2$; $\sin^{-1}(-x)=-\sin^{-1}(x)$; compound identities	DPP 15	45 min
L16	Heights & distances	Angle of elevation/depression; sine rule; cosine rule; 2-observation problems; bearing problems	DPP 16	45 min
L17	Trigonometry PYQ drill I	JEE Main 2019-2022 trig PYQs; identity-based; equation-based; range of trig expressions	DPP 17	45 min
L18	Trigonometry PYQ drill II	JEE Main 2023-2025 trig PYQs; inverse trig; composite functions; heights & distances	DPP 18	45 min

Chapter 3: Complex Numbers | Jun 8 – Jun 15 | 6 Lectures				
L#	**Topic**	**Sub-topics / Key Points to Cover**	**DPP**	**Duration**
L19	Complex number basics	Definition; i and its powers; algebraic form; modulus $\|z\|$; conjugate z^*; argument (principal value)	DPP 19	45 min
L20	Polar form & Euler's formula	$z = r(\cos\theta + i \sin\theta)$; Euler: $e^{i\theta} = \cos\theta + i \sin\theta$; multiplication in polar form	DPP 20	45 min
L21	De Moivre's theorem	Statement & proof; $(r e^{i\theta})^n$; finding powers of complex numbers; roots of unity formula	DPP 21	45 min
L22	Cube roots of unity	$\omega = e^{2\pi i/3}$; properties: $1+\omega+\omega^2=0$; $\omega^3=1$; applications in algebra	DPP 22	45 min
L23	Geometry of complex numbers	Argand plane; locus problems; $\|z-z \| = \|z-z \|$ (perpendicular bisector); $\|z-a\|=r$ (circle); triangles	DPP 23	45 min
L24	Complex numbers PYQ drill	JEE Main 2019-2025 complex number PYQs; locus problems; inequality problems; polynomial with complex roots	DPP 24	45 min

Chapter 4: Quadratic Equations | Jun 16 – Jun 22 | 5 Lectures				
L#	**Topic**	**Sub-topics / Key Points to Cover**	**DPP**	**Duration**
L25	Roots & Vieta's formulas	Discriminant; real/complex/equal roots; sum of roots = $-b/a$; product = c/a; forming equation from roots	DPP 25	45 min
L26	Nature of roots & graph	Both +ve, both −ve, opposite sign, equal; vertex form; maxima/minima; parabola opening direction	DPP 26	45 min
L27	Quadratic inequalities	Sign scheme (wavy curve method); solving $ax^2+bx+c > 0$; inequality with two quadratics	DPP 27	45 min
L28	Common roots & range problems	Condition for common root; both common roots; range of rational function; quadratic identity	DPP 28	45 min
L29	Quadratic PYQ drill	JEE Main 2019-2025 quadratic PYQs; transformation of roots; location of roots; integer roots	DPP 29	45 min

Chapter 5: Sequences & Series | Jun 23 – Jul 1 | 7 Lectures				
L#	**Topic**	**Sub-topics / Key Points to Cover**	**DPP**	**Duration**
L30	AP – nth term & sum	AP definition; $a_n = a + (n-1)d$; $S_n = n/2(2a+(n-1)d)$; properties; AM; inserting AMs	DPP 30	45 min
L31	GP – nth term & sum	GP definition; $a_n = ar^{n-1}$; S_n; infinite GP sum = $a/(1-r)$ for $\|r\|<1$; GM; AM-GM inequality	DPP 31	45 min
L32	HP & AM-GM-HM inequality	HP definition; HM = $2ab/(a+b)$; AM ≥ GM ≥ HM; maximum/minimum using AM-GM; equality condition	DPP 32	45 min
L33	AGP & method of differences	Arithmetico-geometric progression sum; method of differences for S_n; V_n method for telescoping	DPP 33	45 min
L34	Special series	Sum of $1+2+\ldots+n = n(n+1)/2$; $1^2+\ldots+n^2$; $1^3+\ldots+n^3$; problems combining standard series	DPP 34	45 min
L35	Seq & Series PYQ drill	JEE Main 2019-2025 PYQs; sum of n terms; finding missing term; mixed AP+GP problems	DPP 35	45 min
L36	Advanced series problems	Series with partial fractions; series involving combinatorics; series in exponential/log expansion	DPP 36	45 min

Chapter 6: Permutations & Combinations | Jul 2 – Jul 11 | 7 Lectures				
L#	**Topic**	**Sub-topics / Key Points to Cover**	**DPP**	**Duration**

L37	Fundamental counting principle	Multiplication & addition principles; factorial; nPr = n!/(n−r)!; permutations of distinct objects	DPP 37	45 min
L38	Permutations – special cases	Circular permutation (n−1)!; permutations with conditions (specific positions, adjacent/non-adjacent)	DPP 38	45 min
L39	Combinations – nCr	nCr = n!/(r!(n−r)!); properties of nCr; selections with at least/at most; complementary counting	DPP 39	45 min
L40	Distribution problems	Distributing n identical/distinct objects into r groups (with/without restriction); stars & bars	DPP 40	45 min
L41	Derangements & advanced counting	Derangement formula D□ = n!(1−1+1/2!−1/3!+…); inclusion-exclusion principle; multinomial coefficient	DPP 41	45 min
L42	P&C PYQ drill I	JEE Main 2019-2022 P&C PYQs; word problems; rank of word in dictionary; number formation	DPP 42	45 min
L43	P&C PYQ drill II	JEE Main 2023-2025 P&C PYQs; geometry-based counting; graph theory basics; committee selection	DPP 43	45 min

Chapter 7: Binomial Theorem | Jul 13 – Jul 17 | 5 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L44	Binomial theorem & Pascal's triangle	Statement; Pascal's triangle; general term $T_r = {}^nC_r a^{n-r} b^r$; middle term(s)	DPP 44	45 min
L45	Finding specific terms	Term independent of x; coefficient of x^k; term containing x^r; greatest coefficient; greatest term	DPP 45	45 min
L46	Properties & applications	Sum of coefficients (x=1); alternating sum (x=−1); sum of odd/even coefficients; ${}^nC_0 + {}^nC_1 + \ldots = 2^n$	DPP 46	45 min
L47	Multinomial & approximations	Multinomial theorem (brief); binomial approximation $(1+x)^n \approx 1+nx$ for small x; applications	DPP 47	45 min
L48	Binomial PYQ drill	JEE Main 2019-2025 binomial PYQs; term identification; sum of series using binomial; coefficient problems	DPP 48	45 min

Chapter 8: Straight Lines | Jul 27 – Aug 6 | 8 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L49	Distance, section, area formulas	Distance formula; section formula (internal & external); midpoint; centroid; area of triangle; collinearity	DPP 49	45 min
L50	Slope & forms of line	Slope m = tanθ; slope-intercept y=mx+c; point-slope; two-point form; derivation of each	DPP 50	45 min
L51	Intercept & normal forms	Intercept form x/a + y/b = 1; normal form xcosα + ysinα = p; general form ax+by+c=0	DPP 51	45 min
L52	Angle between lines & special lines	$\tan\theta = \lvert(m_1 - m_2)/(1+m_1 m_2)\rvert$; parallel ($m_1 = m_2$); perpendicular ($m_1 m_2 = -1$); angle bisectors	DPP 52	45 min
L53	Distance formulas	Perpendicular distance from point to line; distance between parallel lines; foot of perpendicular; reflection	DPP 53	45 min
L54	Family of lines	$L_1 + \lambda L_2 = 0$ concept; lines through intersection of two lines; locus problems	DPP 54	45 min
L55	Concurrency & locus	Condition for concurrency (3 lines); locus definition; finding locus equation; locus of orthocentre	DPP 55	45 min
L56	Straight lines PYQ drill	JEE Main 2019-2025 straight line PYQs; area of triangle given conditions; midpoint locus; image problems	DPP 56	45 min

Chapter 9: Circles | Aug 6 – Aug 13 | 6 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L57	Equation of circle	Standard $(x-h)^2+(y-k)^2=r^2$; general $x^2+y^2+2gx+2fy+c=0$; centre $(-g,-f)$, $r=\sqrt{(g^2+f^2-c)}$	DPP 57	45 min
L58	Position & tangent	Point inside/on/outside circle; length of tangent; power of point; tangent from external point	DPP 58	45 min
L59	Equation of tangent	Tangent at point on circle (T=0); slope form (y=mx+c with condition); tangent from external point	DPP 59	45 min
L60	Chord & chord of contact	Chord with given midpoint (T=S); chord of contact (T=0 from external point); pole and polar	DPP 60	45 min
L61	Family of circles & radical axis	Radical axis (S −S =0); radical centre; coaxial systems; common chord; orthogonal circles	DPP 61	45 min
L62	Circles PYQ drill	JEE Main 2019-2025 circle PYQs; intercept on axes; locus of midpoint of chord; angle in semicircle	DPP 62	45 min

Chapter 10: Conic Sections | Aug 14 – Sep 1 | 12 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L63	Parabola – standard forms	$y^2=4ax$: focus (a,0), directrix $x=-a$, vertex (0,0), axis y=0, latus rectum length 4a; all 4 forms	DPP 63	45 min
L64	Parabola – parametric & focal chord	Parametric $(at^2, 2at)$; focal chord (t t =−1); semi-latus rectum	DPP 64	45 min
L65	Parabola – tangent & normal	Tangent at t: $ty = x + at^2$; slope form $y = mx + a/m$; normal at t: $y + tx = 2at + at^3$; chord of contact	DPP 65	45 min
L66	Ellipse – standard form	$x^2/a^2 + y^2/b^2 = 1$ (a>b); $b^2 = a^2(1-e^2)$; foci (±ae, 0); directrices; latus rectum $2b^2/a$	DPP 66	45 min
L67	Ellipse – parametric & focal radii	Parametric (acosθ, bsinθ); focal radii r +r = 2a; sum constant property; auxiliary circle	DPP 67	45 min
L68	Ellipse – tangent, normal, director circle	Tangent at (x ,y): xx /a^2+yy /b^2=1; normal; director circle $x^2+y^2=a^2+b^2$; locus of foot	DPP 68	45 min
L69	Hyperbola – standard form	$x^2/a^2 - y^2/b^2 = 1$; $b^2 = a^2(e^2-1)$; foci (±ae, 0); asymptotes y = ±(b/a)x; e > 1	DPP 69	45 min
L70	Hyperbola – conjugate & rectangular	Conjugate hyperbola; combined asymptote equation; rectangular hyperbola $xy = c^2$; parametric (ct, c/t)	DPP 70	45 min
L71	Conics – common properties	Focus-directrix property for all conics; comparison table; identifying conic from general 2nd degree	DPP 71	45 min
L72	Chord of contact & pole-polar	Chord of contact T=0 for all conics; pole-polar relation; pair of tangents SS = T^2	DPP 72	45 min
L73	Conic sections PYQ drill I	JEE Main 2019-2022 conic PYQs; parabola + ellipse problems; locus problems; normals meeting again	DPP 73	45 min
L74	Conic sections PYQ drill II	JEE Main 2023-2025 conic PYQs; hyperbola + combined; conics in general form; hard tangent problems	DPP 74	45 min

Chapter 11: Limits & Derivatives | Sep 2 – Sep 11 | 8 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L75	Limits – concept & standard limits	Left & right limits; existence; standard: sinx/x, (1+x)^(1/x) as x→0; $(1+1/n)^n \to e$	DPP 75	45 min
L76	Limit evaluation methods	Factorization; rationalization; substitution; L'Hôpital's rule; 0/0, ∞/∞ forms	DPP 76	45 min
L77	Continuity	Definition; types of discontinuity (removable, jump, infinite); testing continuity; piecewise functions	DPP 77	45 min

L78	Differentiability	Definition as limit of difference quotient; LHD = RHD; differentiability $\Rightarrow$ continuity; converse false	DPP 78	45 min
L79	Differentiation rules	Sum, product, quotient rules; chain rule; derivatives of all standard functions (trig, inv trig, exp, log)	DPP 79	45 min
L80	Implicit & parametric differentiation	Implicit: differentiate both sides; dy/dx in terms of x & y; parametric: dy/dx = (dy/dt)/(dx/dt)	DPP 80	45 min
L81	Applications: tangent & maxima/minima preview	Slope of tangent = f'(x); normal equation; intro to maxima/minima (critical points); monotonicity intro	DPP 81	45 min
L82	Limits & derivatives PYQ drill	JEE Main 2019-2025 limits & derivatives PYQs; indeterminate forms; chain rule traps; graph-based Qs	DPP 82	45 min

Chapter 12: Statistics | Sep 21 – Sep 24 | 4 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L83	Measures of central tendency	AM for ungrouped & grouped data; weighted mean; median (formula for grouped); mode (modal class)	DPP 83	45 min
L84	Measures of dispersion	Range; mean deviation about mean/median; variance σ^2; standard deviation σ; formula for both	DPP 84	45 min
L85	Combined mean & variance	Combined mean formula; combined variance (parallel axis type formula); coefficient of variation CV = σ/mean	DPP 85	45 min
L86	Statistics PYQ drill	JEE Main 2019-2025 statistics PYQs; change of scale & origin effect on SD/variance; shortcut methods	DPP 86	45 min

Chapter 13: Probability | Sep 25 – Oct 2 | 6 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L87	Classical probability	Sample space; events (simple, compound, mutually exclusive, exhaustive); P(A) = n(A)/n(S)	DPP 87	45 min
L88	Addition & conditional probability	Addition theorem $P(A \cup B) = P(A)+P(B)-P(A\cap B)$; conditional probability $P(A\|B) = P(A\cap B)/P(B)$	DPP 88	45 min
L89	Multiplication theorem	Independent events $P(A\cap B) = P(A)P(B)$; dependent events; probability of independent events in sequence	DPP 89	45 min
L90	Total probability & Bayes'	Law of total probability; Bayes' theorem derivation; posterior probability calculation; examples	DPP 90	45 min
L91	Random variable & expectation	Discrete random variable; PMF; expectation $E(X) = \Sigma x\cdot P(x)$; $E(aX+b)$; variance = $E(X^2)-(E(X))^2$	DPP 91	45 min
L92	Probability PYQ drill	JEE Main 2019-2025 probability PYQs; Bayes' theorem problems; at least/at most probability; Bernoulli trials	DPP 92	45 min

Chapter 14: Mathematical Reasoning | Oct 5 – Oct 7 | 3 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L93	Statements & connectives	Proposition; negation; conjunction (AND); disjunction (OR); conditional (if-then); biconditional (iff)	DPP 93	45 min
L94	Tautology, contradiction & quantifiers	Truth tables; tautology; contradiction; converse, inverse, contrapositive; universal & existential quantifiers	DPP 94	45 min
L95	Reasoning PYQ drill	JEE Main 2019-2025 reasoning PYQs; validating arguments; negation of compound statements	DPP 95	45 min

Phase 5–7: Mathematics Revision & Mock | Oct 22 – Jan 15 | 57 Slots

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
—	Oct 22–Nov 4: Full Math PYQ Sprint + revision	Chapter-wise PYQ drill; Conics hard problems; Calculus applications; Probability Bayes' theorem drill	PYQ sets	45 min
—	Nov 12–Nov 28: Full Revision Cycle 1	Rapid revision; formula derivation check; Conics tangent-normal; Trig equations; P&C advanced problems	Rev DPPs	45 min
—	Nov 29–Dec 19: PYQ Chapter-wise Drill	JEE Main 2019-2025 Math PYQs; Coordinate Geometry (heaviest weight); Sequences; Binomial; Integer type Qs	PYQ sets	45 min
—	Dec 21–Jan 15: Mock Analysis + Weak Ch	Mock analysis; Conics & Limits typically weakest – intensive drill; formula sheet from memory test	Mock papers	45 min